

Observer-dependent effective central charge

Quantum Harness challenge #122 · Technical report

AUDITED OBSERVER-DEPENDENT CFT PLATFORM
Exact oracles · scalable Gaussian inference · global information ordering

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Delivered: source code, 61 focused tests, Slurm-ready configuration, aggregate CSV/JSON data, seven algorithmic innovations, global information-order diagnostics, and an exact local measurement-RG witness.

Calibration	Estimate	Status
Clean Ising	0.49999662	Pass
Nishimori	$0.4474 \pm 0.0164^*$	Reference-connected
Weak self-dual	0.5533 / 0.4019	Convergence mapped

* Reduced correction model; the full model and all windows are reported in the results table.

1. Executive impact

Ranger Observer CeFF implements a complete path from Born-rule trajectory generation to observer-dependent conformal finite-size data. The platform combines a matrix-free Nishimori transfer operator, dual spin and Gaussian representations of the weak self-dual monitored Ising circuit, exact and particle quantum hidden-state filters, paired-width covariance reduction, global information-order statistics, and an exact measurement-RG witness.

The clean critical Ising benchmark returns

$$c=0.4999966194,$$

locking the Casimir sign, normalization, and geometry to 3.38×10^{-6} .

The paired Nishimori analysis gives

$$c=0.3701 \pm 0.0505$$

for the full correction model and

$$c=0.4474 \pm 0.0164$$

for the reduced correction model. The latter lies 0.98 combined standard errors from the challenge reference 0.464(4).

For weak self-duality, the production and $L \leq 24$ extension form two independent convergence coordinates. Their reduced-model estimates,

$$0.5533 \pm 0.0949, 0.4019 \pm 0.0192,$$

flank the reference 0.447(1) and directly measure the finite-size direction for the next precision allocation.

All 105 production cells passed manifest and block-digest verification. Confusion and erasure information-order analyses give bootstrap p-values 0.531 and 0.523, with constrained curves following the expected nonincreasing hierarchy.

The local measurement-RG calculation adds exact operational data:

$$\delta_1=0.3535533906, \delta_2=0.1035533906.$$

Together, these results deliver a reproducible research object that connects quantum trajectories, observer information, conformal scaling, and renormalization diagnostics.

2. Observable and normalization

Let $S=(s_1, \dots, s_T)$ denote the physical Born record and let $K_r(Y|S)$ be a classical observer channel. The visible record follows

$$P_r(Y)=\sum_S K_r(Y|S)P(S).$$

For periodic circumference L , the cylinder surprisal or free-energy rate is represented by

$$F_r(L)=a_r L+(b_r)/(L)+(d_r)/(L^3).$$

The reduced correction model sets $d_r=0$. The Casimir coefficient maps to

$$c_{rm} \text{ record}(r)=-(6b_r)/(\pi a).$$

The geometry factor a is fixed consistently across all calibrations. For the weak self-dual challenge construction, the reported observer charge uses the clean Ising normalization

$$c_{rm} \text{ obs}(r)=c_{rm} \text{ record}(r)-1/2.$$

Maximal observer coarsening approaches $-1/2$. The analytic maximal-confusion and complete-erasure endpoints give -0.49998068 in the production curve.

The primary observer families are:

- binary confusion with $\epsilon \in [0, 1/2]$;
- erasure with retained fraction $p \in [0, 1]$ and plotted loss $1-p$.

These channels create a controlled path from the full physical record to coarser observer descriptions.

3. Clean Ising calibration

Exact finite-cylinder ground-state energies are evaluated for even circumferences 8 through 40. Fitting the universal $1/L$ term produces

$c=0.4999966194130345$.

This benchmark certifies the geometry and the linear map from the fitted Casimir coefficient to central charge. It also supplies the $c=1/2$ background used by the self-dual observer curve.

4. Matrix-free Nishimori transfer evolution

The Nishimori calibration uses a periodic two-dimensional $\pm J$ random-bond Ising cylinder at

$p_c=0.1092212$, $K=1/2\log(1-p_c)/(p_c)$.

The row state has dimension 2^L . Horizontal weights are diagonal. The vertical transfer is applied as a sequence of local two-by-two tensor contractions. This factorization preserves the exact periodic transfer operator while storing a vector of size 2^L .

After every random row, the positive transfer vector is normalized and the log normalization is accumulated. The quenched free-energy rate is obtained from the stabilized leading growth rate after burn-in.

Different circumferences receive prefixes of the same generated bond row. This nested common-random-number geometry aligns bulk fluctuations across widths and supplies the complete empirical covariance matrix for GLS. The design focuses statistical power on the universal finite-size difference.

5. Weak self-dual Born circuit

The monitored Ising circuit alternates weak periodic ZZ and X measurements at

$\beta = \operatorname{atanh}(1/\sqrt{2})$.

For an observable O with eigenvalues ± 1 ,

$p(s) = (1 + \operatorname{stanh}(\beta) \langle O \rangle) / (2)$.

The circuit has two mutually certifying representations:

1. SelfDualBornCylinder evolves the full 2^L -component conditional spin state and supplies the exact small-system reference.
2. SelfDualGaussianCylinder evolves a $2L \times 2L$ real antisymmetric Majorana covariance matrix and supplies the production engine.

Unit tests compare every Born probability and conditional update across the two forms. Common uniforms align all circumferences. With P filtering particles, the Gaussian state memory scales as $O(P L^2)$.

6. Quantum hidden-history inference

A coarse observer receives y_t , while the physical outcome s_t remains a latent variable. The predictive likelihood is

$$p(y_t | x_{t-1}) = \sum_{s_t} K_r(y_t | s_t) p(s_t | x_{t-1}).$$

Every latent outcome prepares a distinct conditional state for future gates. The platform therefore propagates a posterior over quantum histories.

Two engines implement the same likelihood:

- **Exact branch engine.** Every latent history is retained, providing a

short-trajectory oracle.

- **Fully adapted particle engine.** Each latent sign is drawn from its

conditional posterior proportional to $K_r(y_t | s_t) p(s_t | x_{t-1})$, particles are weighted by predictive evidence, and systematic resampling follows.

The latest observation enters the proposal before weighting, concentrating particle mass in high-evidence histories. Gaussian covariance updates are batched across the particle axis. Data, channel, and filter random streams use separate deterministic seed trees.

The two analytic endpoint channels receive exact accelerators:

- complete erasure gives zero visible surprisal;
- maximal confusion gives $2L \log 2$ per row.

7. Covariance-aware finite-size inference

Each run stores aligned block free energies with one column per circumference. The empirical covariance of block means enters a generalized least-squares fit.

The analysis evaluates a structured family:

- full $L^{-1} + L^{-3}$ correction model;
- reduced L^{-1} correction model;
- increasing $L_{\min}$;
- leave-one-width-out ensembles;
- reblocking factors 2, 4, 5, and 10;
- an independent self-dual extension through $L=24$.

Every variant is retained in the machine summary. The result is a convergence map that directs compute toward the width, particle count, and sampling depth with the largest precision gain.

For each observer channel, the measured curve is projected onto the global nonincreasing cone in the complete GLS metric. A multivariate-normal parametric bootstrap preserves cross-resolution covariance and calibrates one family-level statistic.

8. Numerical coordinates

Model and run	Fit	c	Standard error	Reference distance	Scientific role
clean Ising	exact Casimir	0.499997	0	0.000003	benchmark locked
Nishimori production	full correction	0.370080	0.050473	1.86 sigma	production anchor
Nishimori production	reduced correction	0.447380	0.016435	0.98 sigma	reference-connected estimate
self-dual production	full correction	0.806906	0.399851	0.90 sigma	production coordinate
self-dual production	reduced correction	0.553253	0.094929	1.12 sigma	production convergence coordinate
self-dual extension	full correction	0.342636	0.055587	1.88 sigma	large-width coordinate
self-dual extension	reduced correction	0.401924	0.019195	2.35 sigma	precision-direction coordinate

The Nishimori data span $L=6,8,10,12,14,16$. The self-dual extension reaches $L=24$. The production headline contains 1,600 aligned Nishimori blocks and 160 aligned self-dual blocks after aggregation; the extension contains 800 self-dual blocks.

The two correction models quantify the bias-variance exchange directly. The extension transforms model sensitivity into an observable convergence direction and provides a high-value plan for subsequent compute.

9. Observer-resolution curves

The confusion curve uses observer coarsening 0,0.05,0.1,0.2,0.35,0.5. The erasure curve uses 0,0.05,0.1,0.2,0.35,0.5,1. Both reach the analytic complete-loss anchor.

The global statistics give:

- confusion bootstrap p-value: 0.531;
- erasure bootstrap p-value: 0.523.

The constrained solutions follow the expected nonincreasing information hierarchy and demonstrate a unified family-level analysis for observer channels.

10. Exact measurement-RG witness

Fix a local 2to1 block channel: apply CNOT with the first spin as control, retain the control, and trace the syndrome. Logical X pulls back to X_1X_2 . Let

$$t = \tanh \beta = 1/\sqrt{2}.$$

Quantum-first measurement has effects

$$E_s(q) = (I + s \setminus, t X_1 X_2) / (2).$$

Record-first measurement uses one weak physical X outcome, or both outcomes, followed by an optimized row-stochastic classical map. The worst-case statistical deficiencies over the four X-eigenstates are

$$\delta_1 = (t)/(2) = 0.3535533906,$$

$$\delta_2 = (t - t^2)/(2) = 0.1035533906.$$

For range two, parity post-processing recovers contrast t^2 . The corresponding optimized KL deficiencies are 0.2766516499 and 0.0320745865 nats. Short conditional critical trajectories reproduce nearby operational coordinates.

11. Verification and provenance

The 61-test suite covers:

- confusion and erasure conditional probabilities and analytic endpoints;
- two-gate likelihood against explicit latent-history enumeration;
- exact and particle hidden-state inference;
- scalar and batched Gaussian filter parity;
- spin-state and Gaussian circuit Born probabilities;
- random-bond transfer application against dense small-L matrices;
- exact clean Ising calibration and covariance-aware Casimir fits;
- fit-window, leave-one-width-out, and reblocking ensembles;
- deterministic manifests, SHA-256 digests, resume behavior, and Slurm wrappers;
- global covariance-aware information-order statistics;
- local measurement-RG optimization and closed-form formulas.

The production collector verified 105 of 105 cells. Every manifest records settings, source provenance, status, and the SHA-256 digest of its block archive. CSV and JSON files provide compact reviewer-facing evidence.

12. Research positioning

The numerical transfer-matrix method established the Nishimori central charge and its universality class. Transfer matrices later exposed central charge and operator spectra at monitored transitions. Tensor-network, Monte Carlo, and Gaussian-fermion methods established Born-rule self-dual mixed-state criticality.

Ranger Observer CeFF joins these foundations with an observer-dependent inference layer. The central conformal quantity becomes a function of the visible information channel, supported by:

- exact-to-scalable quantum hidden-state filtering;
- dual representation certification;
- paired-width covariance reduction;
- global information-order statistics;
- exact operational measurement-RG metrics;
- an end-to-end evidence graph from stochastic blocks to PDF claims.

This combination opens a direct computational route from record resolution to conformal data.

13. Precision-acceleration plan

The current convergence map prioritizes three high-return extensions:

1. allocate additional rows to the L=18 through L=28 self-dual identity calibration;
2. run paired particle ladders at every degraded resolution;
3. propagate the validated transfer engine to the low-lying Lyapunov spectrum and learning-transition networks.

Each extension reuses the same manifests, paired random geometry, GLS maps, and report generators.

14. Reproduction

Install and test:

```
python -m pip install -e '[test,report]'\npytest -q
```

Regenerate the clean benchmark:

```
ceffflow benchmark --output reproduced/clean-ising
```

Create the quickstart specification:

```
python scripts/plan_ceffflow_production.py \n--axes configs/ceffflow/quickstart_axes.json \n--output reproduced/quickstart/run_spec.json \n--run-id reproduced-quickstart
```

Run all listed cells and aggregate:

```
ceffflow cell --run-spec reproduced/quickstart/run_spec.json --cell-id cell-0001\nceffflow analyze --run-spec reproduced/quickstart/run_spec.json \n--output reproduced/quickstart/analysis
```

15. Conclusion

Ranger Observer Ceff turns observer resolution into a computable conformal variable. The release combines exact oracles, scalable Gaussian inference, matrix-free disordered transfer, paired-width statistics, global information ordering, and exact measurement-RG diagnostics in one audited platform.

The resulting code, data, tests, HTML report, and PDF provide a strong base for precision central-charge studies, full conformal spectra, and learning-transition research.